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<https://github.com/c-nos>

SKILLS

Python (Numpy, Pandas, Matplotlib, Seaborn, Tensorflow, PyTorch, Keras), Linux, Machine Learning, A/B Testing, Hypothesis Testing, Technical Writing, Microsoft suite (Word, Excel, Visio, and Powerpoint), Mathematica, Lyx, Quantum Computing, Condensed Matter Physics.

PROFILE

I am eager to leverage my background in physics and data science to tackle challenging R&D problems. My most recent experience in the Sensors group at PSI Inc. included developing algorithms and data pipelines for diverse sensing modalities, including RF data, chemical data, and multimodal geospatial data, among others. With an educational background in condensed matter physics from UC Berkeley and the Weizmann Institute, combined with data science experience from PSI Inc. and BrainStation, I am well-equipped to contribute to innovative projects. I am now seeking exciting new R&D opportunities where I can apply my skills and contribute value in a dynamic and cutting-edge environment.

EDUCATION

BrainStation | Data Science

APR 2022 – JUL 2022

Weizmann Institute of Science | M.Sc. in Physics

AUG 2014 – MAY 2017

University of California, Berkeley | B.A. (Honors) In Physics

AUG 2010 – MAY 2014

Casey Nosiglia

COMPUTATIONAL PHYSICIST

EXPERIENCE

Project Analyst | Physical Sciences Incorporated

Oct 2023 – MAR 2025, ANDOVER, MASSACHUSETTS, USA

- Developed statistical methods for chemical detection in cluttered environments from mass spectrometer data under the DARPA SIGMA+ program; achieved 90% detection accuracy at below ppb sensitivity for a select subset of chemicals.
- Deployed the above algorithm to run on embedded hardware, which continuously communicates with the mass spectrometer for real-time chemical detection.
- Trained an ML model to quantify the hardness of concrete from RF antenna array measurements. Improved accuracy by an order of magnitude from initial results, satisfying accuracy KPIs.
- Submitted two \$200k+ proposals for government contracts (SBIR & STTR).

Technical Specialist | McCoy Russell LLP

DEC 2020 - MAR 2022, PORTLAND, OREGON, USA

- Drafted 24 patent applications across various technology sectors, including internal combustion engines, electric motors, medical devices, and optics.
- Developed strong technical writing, research, and collaboration skills, while guiding invention patents from ideation to USPTO submission.

Research Assistant | Weizmann Institute of Science

JAN 2015 – AUG 2017, REHOVOT, ISRAEL

- Modelled thermal and electrical transport properties of edge currents in Fractional Quantum Hall Effect (FQHE) systems using differential equations, applying analytical and numerical methods in Mathematica.

PROJECTS

Data Scientist | Identifying Plant Diseases in Leaf Images – Capstone

JUN 2022 – JUL 2022, BrainStation

- Used Python, Tensorflow, Keras, and related libraries to prepare and process leaf image data, in order to identify plant health (healthy/diseased) and disease type among 14 different plant species.
- Employed transfer learning with the ResNet50 CNN model in order to identify plant health and disease type among the different plant species with high accuracy (~90% in both cases).

Main Researcher | M.Sc. Thesis Project

JAN 2015 – AUG 2017, Weizmann Institute of Science, Rehovot, Israel

- Modelled electrical and thermal transport properties of edge currents in FQHE systems, in order to disambiguate competing models of the edge currents.
- Applied physics modelling to derive a set of differential equations, and used both analytical and numerical solutions (with Mathematica) to extract key behaviors of competing models.
- Visualized results and proposed experimental settings to distinguish competing models; wrote and submitted paper to *Physical Review B* (accepted).